INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE NAME	

CLASS		INDEX NUMBER	

BIOLOGY 9747/01

Paper 1 Multiple Choice **24 September 2009**

1 hour 15 min

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.

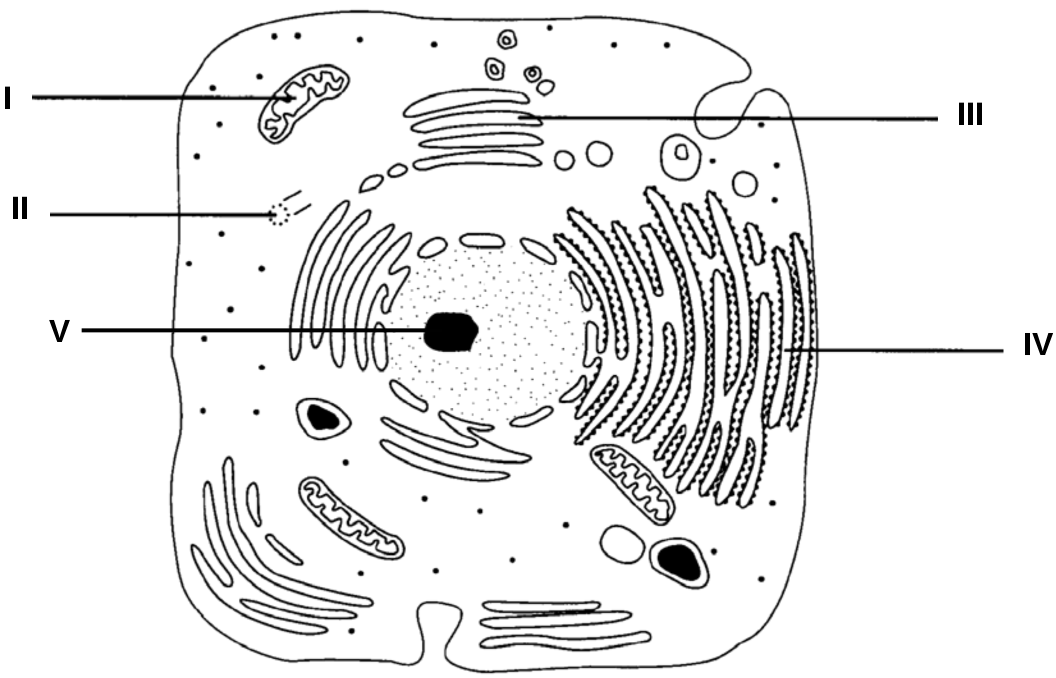
There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
Calculators may be used.

This document consists of **17** printed pages.

1 The diagram shows the ultrastructure of a cell.



Which of the following structure(s) contain(s) nucleic acids?

A I and II only
B I and III only
C I and IV only
D I, IV and V only

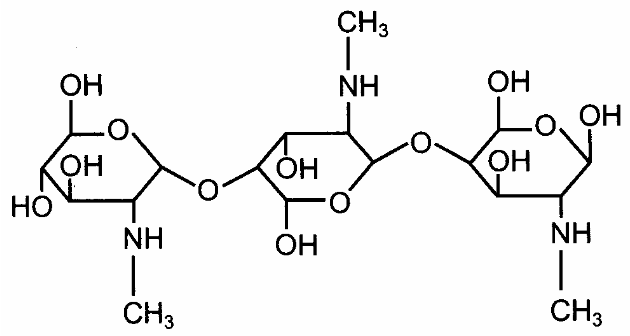
2 The diagram shows how the enzyme glutamine synthetase removes the ammonia produced during plant metabolism.

$$\text{ammonia + glutamate} \xrightarrow{\text{glutamine synthetase}} \text{glutamine}$$

Some herbicides contain an active agent which resembles glutamate.
What is the likely mode of action of this agent?

A It acts as an end-product inhibitor.
B It acts as a competitive inhibitor.
C It decreases levels of ammonia.
D It increases levels of glutamate.

3 The diagram shows the structure of a competitive inhibitor of the enzyme lysozyme.



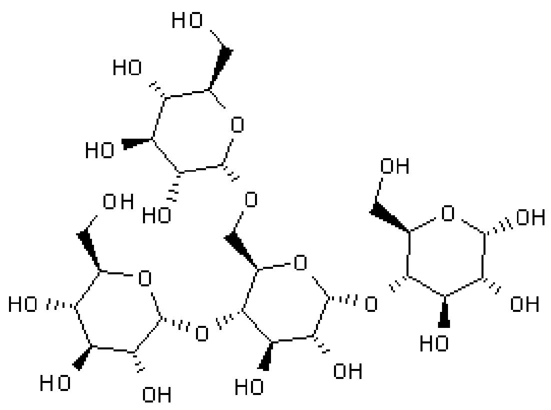
Which substance is most likely to be the normal substrate of lysozyme?

A Lipid
B Polynucleotide
C Polypeptide
D Polysaccharide

4 Which solution contains high concentration of sucrose and low concentration of glucose?

Solution	*Mass of red ppt from Benedict's test / mg*	
	without acid hydrolysis	with acid hydrolysis
A	0.5	2.5
B	2.5	2.5
C	1.3	1.8
D	0	3.0

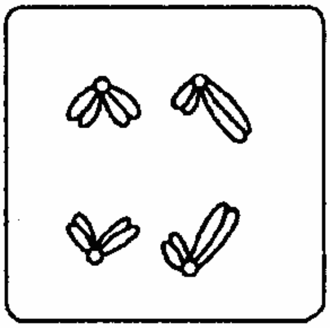
5 The diagram shows a macromolecule found in animals. Which of the following identifies it?



A Cellulose
B Collagen
C DNA
D Glycogen

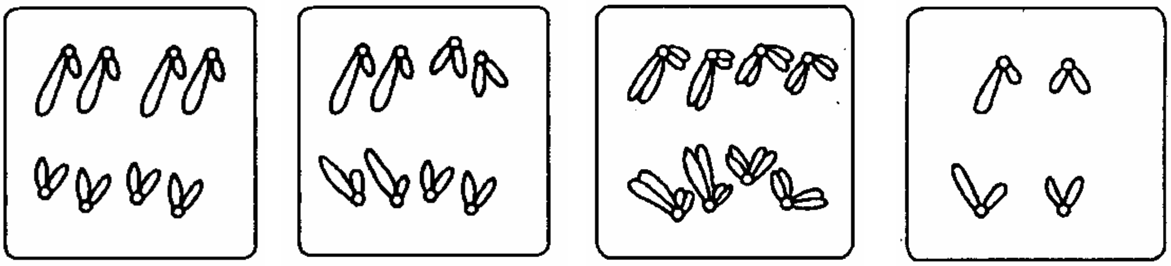
 [Turn over

6 The following diagram shows a cell at anaphase I of meiosis.



Which diagram shows anaphase during mitosis in the same organism?

A **B** **C** **D**



7 Which sugar and base, in addition to inorganic phosphate, will be released from the hydrolysis of a certain pyrimidine nucleotide?

	sugar	*base*
A	deoxyribose	uracil
B	deoxyribose	guanine
C	pentose	uracil
D	ribose	adenine

8 Cytosine comprises 0.32 of the nitrogenous bases in a sample of DNA consisting of 200 base pairs.

What is the total number of adenine and thymine bases in a single strand of DNA from this sample?

A 16
B 18
C 36
D 72

9 The table shows DNA triplets and their corresponding amino acids.

DNA triplet	AGG	ATG	CGC	CGT	GCA	GCG	TAC	TCC	TGG
amino acid	ser	tyr	ala	ala	arg	arg	met	arg	thr

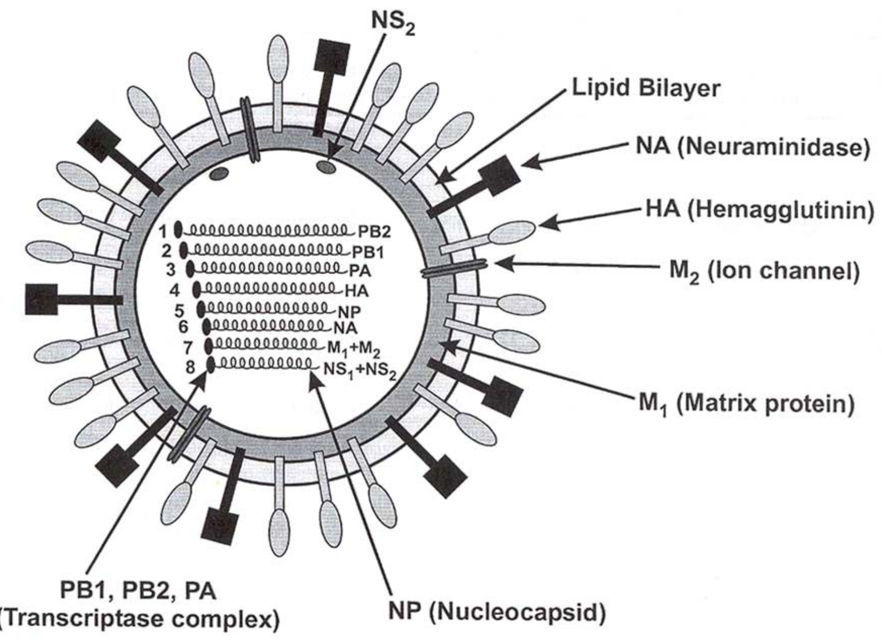
Part of the mRNA sequence produced following transcription of a gene is shown.

GCAGCGUACUCC

What is the correct sequence of amino acids produced when this mRNA is translated?

A ala-ala-thr-ser
B ala-ala-tyr-ser
C arg-arg-met-arg
D arg-arg-met-thr

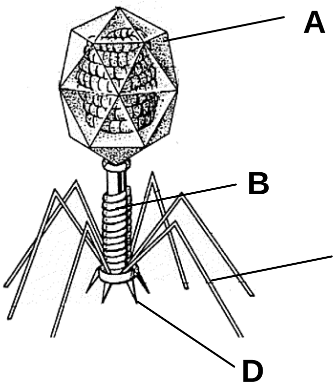
10 The diagram shows an influenza A virus.



Mutation in which of the following will affect the ability of the virus to replicate its genome in the host cell?

A HA
B NA
C NS1 + NS2
D PB1, PB2, PA

11 The diagram shows a bacteriophage.

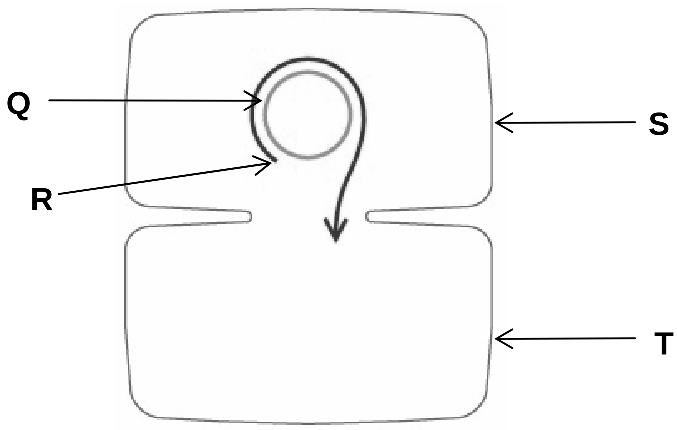


Which structure plays the most important role in host recognition?

12 Bacterial Strain A has been infected with viruses. Upon lysis, the virions produced are introduced to Bacterial Strain B. After a short period of time, a new strain of bacteria is detected that is very similar to Strain A but has a few characteristics of Strain B. Which is the process that leads to the production of the new strain?

A transduction
B transposition
C transformation
D conjugation

13 The diagram shows two bacteria undergoing conjugation.



Which of the following identifies the structures labelled?

	Q	R	S	T
A	chromosome	origin of replication	F^+ cell	F^- cell
B	plasmid	3' end	Hfr cell	F^- cell
C	plasmid	origin of transfer	F^+ cell	F^- cell
D	recombinant plasmid	plasmid	F^+ cell	Hfr cell

14 The diagram shows the lac operon.

P_I	*LacI*		P_{lac}	O	*LacZ*	*LacY*	*LacA*

Mutation in which of the following DNA sequence(s) would result in the constitutive synthesis of β-galactosidase?

A *LacZ* and *Lac Y*
B *LacI* and P_{lac}
C O and *LacI*
D P_I and P_{lac}

15 Which of the following **does not** describe the function of telomeres?

A maintain integrity of genes
B prevent cancerous cell growth
C prevent chromosome fusion
D regulate gene transcription

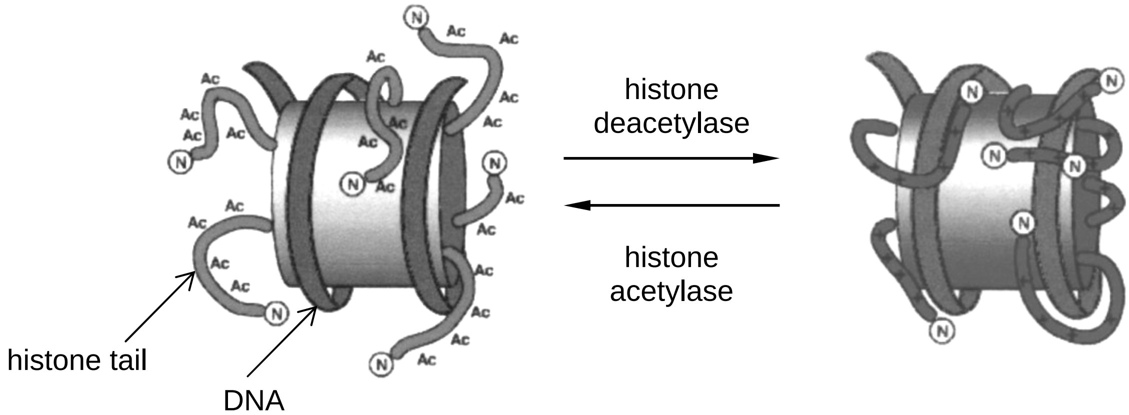
16 Which of the following is **not** a form of gene regulation in eukaryotes?

A chromosome modification
B inducible operons
C interaction of control elements
D RNA splicing

17 Post-translational control in eukaryotic cells can be done by

A inactivating essential transcription factors
B preventing attachment of ribosomes to mRNA
C selectively degrading the mRNA transcript
D selectively ubiquitinating proteins

18 The diagram below shows how acetylation of histones promotes loose chromatin structure. Recent evidence has shown that chemical modification of histones play a direct role in regulation of gene expression. Which of the following best explains how acetylation regulates gene expression?



A Helicase action is enhanced by acetylation.
B Acetylation of histones neutralizes their negative charges and encourages binding to DNA polymerase.
C When nucleosomes are highly acetylated, chromatin becomes less compact and DNA is more accessible for transcription.
D RNA polymerase works better by binding with acetyl groups.

19 Which of the following shows the order in which the structures would be used in the process of gene expression resulting in the formation of a protein?

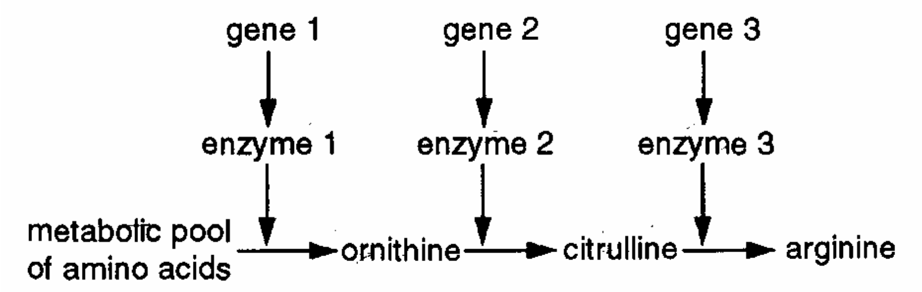
	earliest →			***latest***
A	polyribosome	promoter	spliceosome	primary transcript
B	spliceosome	promoter	primary transcript	polyribosome
C	promoter	primary transcript	spliceosome	polyribosome
D	primary transcript	spliceosome	promoter	polyribosome

20 In the fruit fly, *Drosophila melanogaster*, the diploid number of chromosomes is 8.

In the absence of crossing over or mutation, how many genetically unique kinds of gamete might be formed by one individual?

A 4
B 8
C 16
D 32

21 The diagram shows a metabolic pathway involving some amino acids.



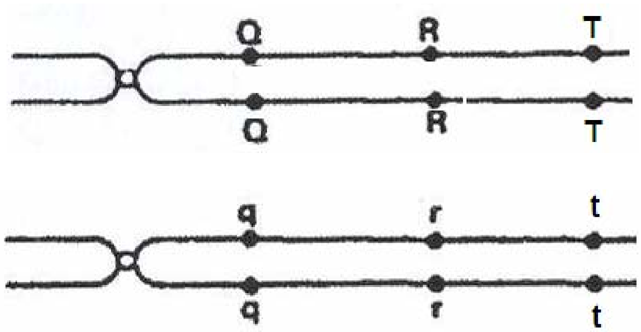
A strain of *Escherichia coli* possesses a mutant form of gene 2. What should be supplied to the culture medium to enable growth of this mutant strain?

A citrulline
B ornithine
C arginine and citrulline
D citrulline and ornithine

22 Which of the following mutation is most likely to result in aneuploidy?

A anaphase promoting complex genes
B growth hormone genes
C p53 gene
D ras gene

23 The diagram shows bivalent during prophase I of meiosis.



How many chiasma(ta) must be formed to obtain a gamete with the genotype **qRt**?

A 0 **B** 1 **C** 2 **D** 3

[Turn over

24 In a small mammal, the allele for grey fur, **G**, is dominant to that for white fur, **g**. The allele for long tail, **T**, is dominant to the allele for short tail, **t**. Animals with grey fur and long tails were crossed with those having white fur and short tails.

The table shows the phenotypes of the 55 offspring.

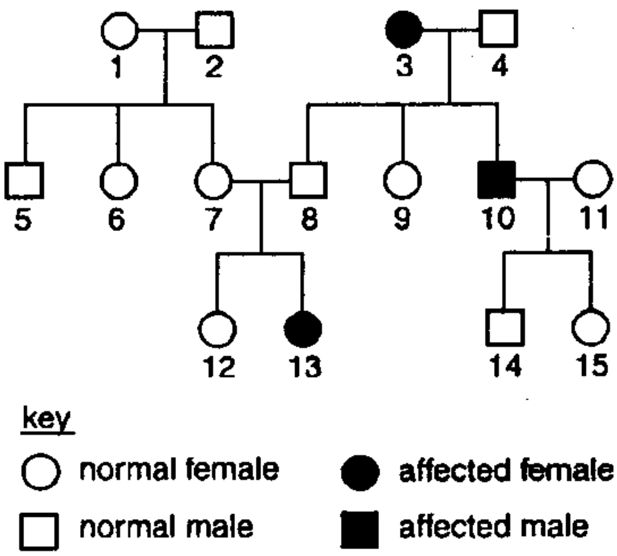
Number of offspring	***fur***	***tail***
15	grey	long
14	grey	short
14	white	long
12	white	short

What were the genotypes of the parents?

A Ggtt x ggtt
B GGTt x Ggtt
C GgTt x GgTt
D GgTt x ggtt

Refer to the following family tree for Questions **25** and **26**.

25 The family tree shows the inheritance of a genetic condition governed by the R/r gene.



Which of the females are certain to have the genotype Rr?

A **1, 6** and **7**
B **1, 7** and **12**
C **7, 9** and **15**
D **9, 12** and **15**

26 What is the probability of **13** and **14** having an affected daughter?

A 0
B 0.25
C 0.5
D 0.75

27 Which process is an example of active transport?

A Movement of potassium ions out of a neurone at resting potential.
B Movement of protons through the stalked particle.
C Movement of sodium ions into a neurone during generation of action potential.
D Release of acetylcholine into the synaptic cleft.

28 It has been found that an aqueous suspension of isolated chloroplasts will evolve oxygen if illuminated in the presence of a certain type of compound.

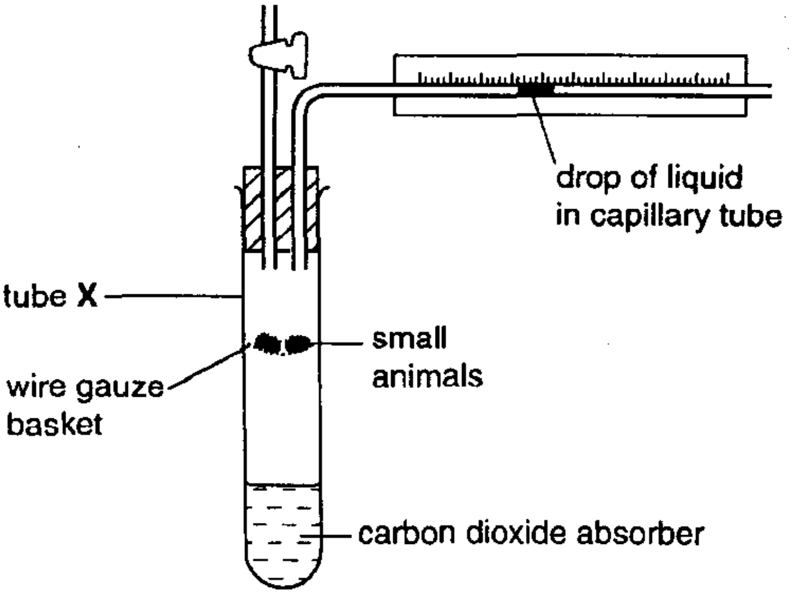
Which type of compound and which wavelength of light would give maximum oxygen evolution?

	type of compound	***wavelength of light at which maximum evolution occurs / nm***
A	electron acceptor	300 and 550
B	electron acceptor	450 and 700
C	electron donor	680 and 700
D	electron donor	420 and 600

29 Which of the following is **true** about Calvin cycle?

A Its cessation will cause light dependent reactions to stop.
B It can only take place under dark conditions.
C It occurs in the thylakoid space.
D Its progression is independent of the light dependent reactions.

30 The diagram shows a simple respirometer.



The changes in gas volume in the tube are measured at intervals.

time (minutes)	gas volume with carbon dioxide absorber (cm^3)	gas volume without carbon dioxide absorber (cm^3)
0	0.0	0.0
10	–0.4	–0.1
20	–0.8	–0.2
30	–1.2	–0.3

Tube **X** contains 2 g of small animals. What is the carbon dioxide output per g per hour for these organisms?

A 0.9 cm^3 **B** 1.8 cm^3 **C** 2.4 cm^3 **D** 4.8 cm^3

31 When the concentration of glucose in the blood rises above the normal range, the pancreas releases insulin. Some humans cannot make enough insulin and must have regular insulin injections. Different forms of insulin have been produced by pharmaceutical companies. The table below shows the effects of three forms of insulin.

Different forms of insulin	Onset of action	Peak action	Duration
P	15 min	30 to 90 min	3 to 5 hours
Q	30 min	2 to 4 hours	6 to 8 hours
R	4 to 8 hours	12 to 18 hours	24 to 28 hours

Which of the following conclusion corresponds best to the information shown?

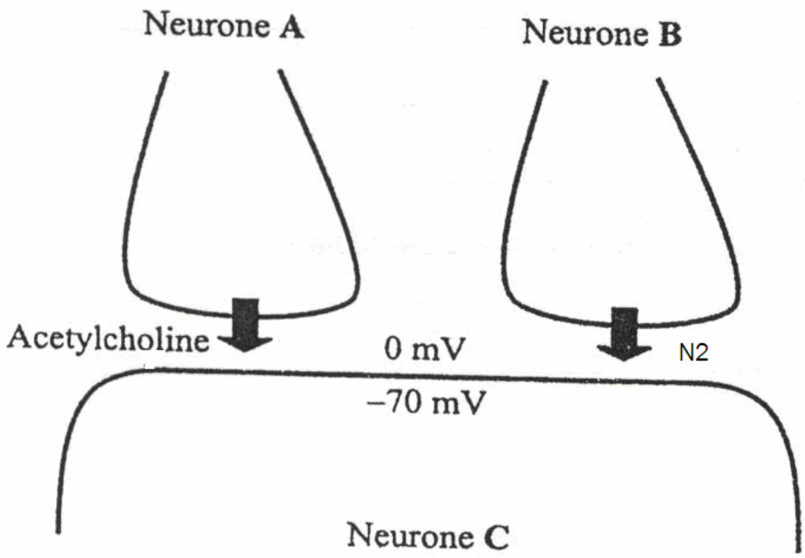
A Form **P** is good because a high blood glucose level may be reduced quickly.

B Form **Q** has a longer duration of action than the other two forms, which means that a diabetic may need fewer injections.

C Form **Q** is good because a low blood glucose level can be raised quickly as it lasts longer than Form **P**.

D Form **R** has a longer duration than the other two forms, which means that a diabetic can consume more glucose per meal.

32 The diagram below shows axon terminals from each of two neurones, **A** and **B**, forming synapses with a third neurone, **C**. When stimulated, neurone **A** releases acetylcholine while neurone **B** releases a structurally different neurotransmitter, N2. Concentrations of some ions are also shown.



	Ion concentration / mmoldm^{-3}		
	Na^+	**K^+**	**Cl^-**
Outside neurone C	145	12	120
Inside neurone C	12	155	5

Which of the following statements best explains why an action potential is not triggered in neurone **C** when both neurones **A** and **B** are stimulated?

A N2 competes with acetylcholine for receptor binding at the postsynaptic membrane of neurone C, resulting in less Na^+ ions entering to cause depolarization.

B N2 causes the membrane potential of neurone **C** to be less negative, therefore more acetylcholine must be released by neurone **A** to cause greater influx of Na^+ ions.

C N2 increases the permeability of the postsynaptic membrane of neurone **C** to Cl^- ions, resulting in hyperpolarization.

D Acetylcholine is released slowly from neurone **A** while N2 is released rapidly from neurone **B**.

33 The chart shows the classification of two species of crocodile.

Animalia
Chordata
Reptilia
Loricata
Crocodylidae
Crocodylus
Crocodylus niloticus Nile crocodile
Crocodylus porosus Salt water crocodile

To which family do these crocodiles belong?

A Chordata
B Crocodylidae
C Loricata
D Reptilia

34 In the summer of 1995, at least 15 iguanas survived Hurricane Marilyn on a raft of uprooted trees. They rode the high seas for a month before colonizing the Caribbean island, Anguilla. These few individuals were perhaps the first of their species, *Iguana iguana*, to reach the island. When one such healthy iguana was mated with another from its source of origin in the Virgin Islands, the offspring were infertile.

Which of the following best explains this?

A Founder effect
B Artificial selection
C Allopatric speciation
D Diversifying selection

35 Which of the following ideas were **not** parts of Darwin's theory of evolution by natural selection?

I Acquired variation is inherited

II Variation between individuals arises by gene mutation

III Organisms produce more offspring than the environment can support

IV Only those individuals best adapted to the environment survive and reproduce

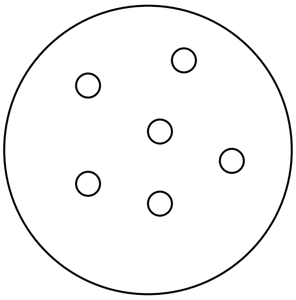
V Regions that encode portions of the polypeptide that are vital for structure and function are less likely to incur mutations

A I, II and V
B I, III and IV
C II, III and V
D II, IV and V

36 Which of the following statements is **true**?

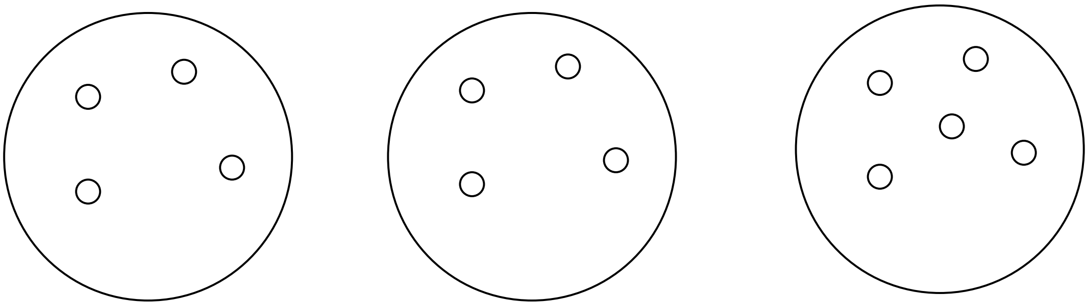
A Evolution does not occur in the absence of mutation.
B Genetic drift occurs when a few individuals migrate from one population to another.
C Natural selection speeds up the rate of mutation in a population.
D Silent mutations are selectively neutral.

37 A mixture of insulin gene, plasmid with ampicilin and tetracycline resistant genes, *EcoR*I and DNA ligase was allowed to react and subsequently heat shocked with *E. coli*. The mixture was then plated on nutrient agar to obtain a master plate with colonies shown below.



Master plate (nutrient agar)

Three replica plates containing different antibiotics were obtained from the master plate. Diagram below shows the results of the replica plates.



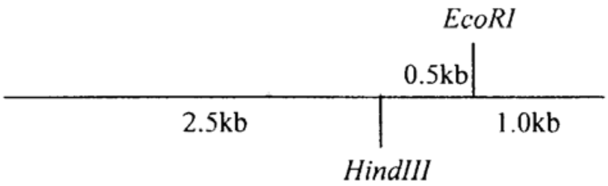
Nutrient agar with ampicillin | Nutrient agar with ampicillin and tetracycline | Nutrient agar with tetracycline

Which of the following can be concluded from the results obtained?

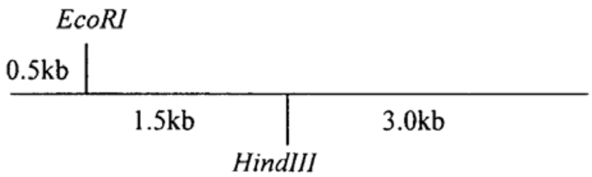
A There is one untransformed colony.
B There are two colonies transformed with recombinant plasmid.
C The tetracycline resistant gene is inactivated by insertion of the insulin gene.
D The ampicillin resistant gene does not act as a selectable marker.

38 Digestion of a 4 kb DNA molecule with *EcoR*I yields two fragments of 1 kb and 3 kb each. Digestion of the same molecule with *Hind*III yields fragments of 1.5 kb and 2.5 kb. Finally, digestion with *EcoR*I and *Hind*III in combination yields fragments of 0.5 kb, 1 kb and 2.5 kb. How would a restriction map indicating the positions of the *EcoR*I and *Hind*III cleavage sites look like?

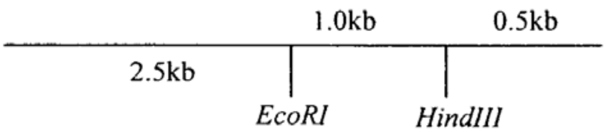
A



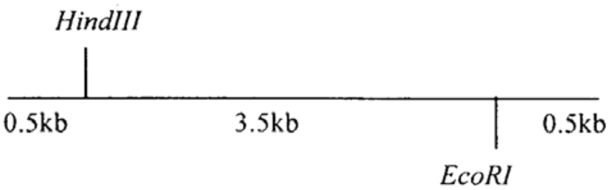
B



C



D



39 Where in the human body do stem cells giving rise to lymphocytes divide?

A blood stream
B bone marrow
C liver
D lymph nodes

40 Maize varieties are being developed in which the leaves produce proteins that are toxic to insects. The DNA coding for these toxic proteins was inserted into a maize chromosome via a bacterial plasmid. Many people are opposed to this process.

Which objection is **not** biologically valid?

A Beneficial insects may be killed if they eat genetically modified maize.
B Genes for antibiotic resistance are present in plasmids and these genes may pass to harmful bacteria.
C Hybridisation may transfer the bacterial genes from maize to weeds, giving the weed species new and harmful characteristics.
D Mutations may be caused in cattle or humans that eat the genetically modified maize.